STUDY MATERIAL: TYPES OF ARTIFICIAL INTELLIGENCE

First-Year B.Tech | Simple Notes for Exams

1. Introduction to Artificial Intelligence (AI)

Artificial Intelligence (AI) is a branch of computer science that enables machines and computers to perform tasks that normally require human intelligence.

These tasks include learning, problem-solving, understanding language, recognizing images, and making decisions.

AI can be classified into different types based on its capabilities. Three commonly discussed categories are Narrow AI, Broad AI, and General AI.

2. Narrow AI (Weak AI)

Definition: Narrow AI is an artificial intelligence system designed to perform one specific task or a limited set of related tasks.

It works well within its assigned purpose but cannot automatically perform every other task like a human.

Real-world examples:

1. **Google Maps:** Uses AI to predict traffic, suggest routes, and estimate travel time.
2. **Face recognition:** Smartphones use AI to recognize a user's face and unlock the device.

Advantages:

- Performs specific tasks quickly and accurately.
- Saves time and human effort.
- Can process large amounts of data.

Limitations:

- Has limited capabilities outside its designed task.
- Does not possess general human-like intelligence.

Key point: Narrow AI is designed for a specific purpose and is widely used today.

3. Broad AI

Definition: Broad AI refers to AI systems that can work across multiple different tasks or areas, rather than being restricted to just one narrow task. However, they may still have limitations and do not necessarily possess human-level intelligence in every area.

The term “Broad AI” is not used as consistently as “Narrow AI” and “Artificial General Intelligence (AGI)” in technical literature.

Real-world examples:

1. **ChatGPT:** Can help explain subjects, write content, summarize notes, answer questions, and assist with coding.
2. **Google Gemini:** Can assist with writing, answering questions, understanding images, and solving different kinds of problems, depending on the version and features available.

Advantages:

- Supports a variety of tasks.
- Can help users with learning, writing, and problem-solving.
- Reduces the need to use separate tools for every task.

Limitations:

- May provide incorrect information.
- Can struggle with unfamiliar or complex tasks.
- Its abilities depend on the model, available tools, and context.

Key point: Broad AI can support multiple kinds of tasks, but this does not mean it has human-level general intelligence.

4. General AI (Artificial General Intelligence — AGI)

Definition: General AI is a hypothetical form of AI that would be able to learn, understand, reason, and solve problems across a wide range of tasks at a human-like level.

Unlike Narrow AI, it would not be limited to one particular task. It would be able to apply knowledge learned in one area to new situations in other areas.

Real-world examples and illustrations:

1. **Human-like intelligent robot (hypothetical):** A robot that could learn to cook, repair a bicycle, teach a student, and adapt to completely new tasks without being separately programmed for each one.
2. **An adaptable virtual assistant (hypothetical):** An AI assistant that could learn new skills, understand unfamiliar situations, plan complex activities, and transfer its knowledge between subjects as flexibly as a human.

Important note: These are hypothetical illustrations, not confirmed examples of existing AGI. As of 2026, there is no generally accepted demonstration of AI achieving full human-level general intelligence across all domains.

Potential advantages:

- Could solve a wide variety of unfamiliar problems.
- Could transfer knowledge between different fields.
- Could assist with complex scientific and everyday tasks.

Challenges and concerns:

- Building reliable AGI is a major scientific challenge.
- Safety, privacy, control, and ethical issues must be addressed.
- Its full capabilities and possible effects remain uncertain.

Key point: General AI aims for flexible, human-like intelligence across many different tasks and remains a research goal.

5. Difference Between Narrow AI, Broad AI, and General AI

Feature	Narrow AI	Broad AI	General AI
Main ability	Specific task	Multiple tasks	General human-like ability
Flexibility	Limited	Wider range	Highly adaptable in principle
Example	Face recognition	ChatGPT	Hypothetical intelligent robot
Current status	Widely available	Available in various forms	Not yet achieved in the full AGI sense
Learning across fields	Very limited	Possible, with limitations	Expected to be highly flexible

6. Quick Revision

- **Narrow AI:** Designed to perform a specific task.
- **Broad AI:** Can assist with several different tasks.
- **General AI:** A proposed AI that could learn and solve a wide range of problems at a human-like level.

Easy way to remember

Think of three students:

- **Narrow AI:** A student who is excellent at solving only mathematics problems.
- **Broad AI:** A student who can help with mathematics, science, writing, and coding, but still has limitations.

- **General AI:** A hypothetical student who can learn almost any subject, adapt to new situations, and apply knowledge flexibly like a human.

7. Important Exam Questions

1. Define Artificial Intelligence.
2. What is Narrow AI? Give two examples.
3. Explain Broad AI with two examples.
4. What is General AI? Why is it considered a research goal?
5. Differentiate between Narrow AI, Broad AI, and General AI.
6. Write two advantages and two limitations of Narrow AI.

Conclusion

Artificial Intelligence is transforming many areas of modern life. Narrow AI performs specific tasks, Broad AI supports multiple types of tasks, and General AI represents the goal of developing flexible, human-like intelligence. Understanding these categories helps us understand both the capabilities of today's AI systems and the goals of future AI research.